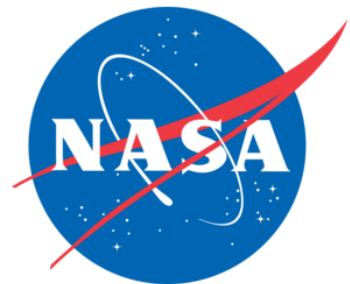


NOVA CORP

NOVA PROJECT

Members:

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Objectives

We have developed a web platform whose primary goal is to enable seamless zooming in and out of large-scale astronomical images without loss of quality. Additionally, the platform integrates an automated analysis system that classifies stars in the images based on their color, allowing users to infer key physical properties such as temperature, spectral type, and evolutionary stage.

This combination of high-fidelity visualization and real-time analysis enables a deeper understanding of stellar environments, supporting both scientific research and public astronomy outreach, while laying the groundwork for future applications in planetary exploration and cosmic observation.

Project Development

The first step was to create a website for this project. We chose the Flask web framework due to its simplicity and beginner-friendly nature.

We then installed the necessary Python libraries and created a project folder containing the core files:

- nova_modul.py (for custom functions and analysis),
- app.py (the main Flask application),
- index.html (the frontend interface).

Next, we set up a virtual environment via the terminal to isolate project dependencies. Once activated, the virtual environment allowed us to launch a local web server (localhost), enabling us to test the application on our own machines.

To share the prototype with team members and external users, we used ngrok. After downloading the ngrok executable and running it on the required port, we generated a public URL that made the locally hosted site accessible over the internet.

At this stage, the project remains a proof-of-concept prototype, demonstrating how the final platform could look and function.

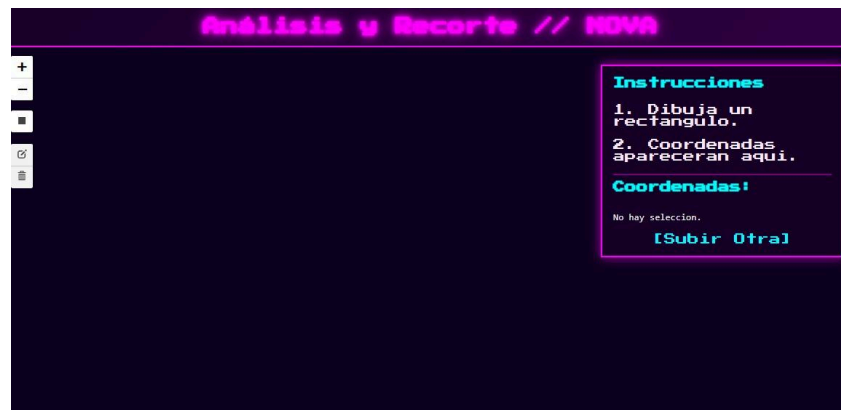


Image processing was performed to identify the type and number of stars in the image by applying filters to separate them. Additionally, code was developed to enable zoom functionality on the previously processed images and to integrate this capability with the web platform, allowing for more efficient rendering and seamless compilation of the images.

vision and mission

We are committed to continuously enhancing the platform to deliver an even richer, more precise, and interactive user experience. This includes:

- Expanding detection and analysis capabilities beyond stars to include galaxies, nebulae, planets, asteroids, and other celestial objects.
- Enriching the data with detailed astrophysical information such as temperature, luminosity, distance, estimated mass, spectral type, variability, proper motion, and more.
- Implementing interactive tools that allow users to compare multi-spectral images (optical, infrared, ultraviolet) and perform simple measurements (e.g., angular distances, relative brightness).
- Incorporating collaborative features, such as the ability to tag objects, report anomalies, and contribute to citizen science initiatives.

Our ultimate goal is to transform this platform into a dynamic space for exploration, learning, and discovery—where anyone, from a student to a professional researcher, can engage with the universe in an intuitive, rigorous, and meaningful way.

conclusion

This project holds significant value for understanding the environments captured in astronomical imagery. By enabling deep zooming without substantial loss of image quality, it not only facilitates detailed observation of space but also makes it possible to identify regions of interest on planets, moons, and other celestial bodies.

These capabilities are essential for current scientific research and could prove critical in planning future space exploration missions—helping to detect geological features, anomalies, potential landing sites, or even signs of dynamic processes in near real time. Ultimately, the platform does more than democratize access to high-resolution astronomical data; it becomes a strategic tool for discovery and decision-making in space exploration.